6CCSAALA: Advanced Logics for Artificial Intelligence

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8 Łukasiewicz Logic and McNaughton's Theorem

Learning Objectives

By the end of this section you should be able to:

- ☐ Show that the truth-function of every propositional formula in 1 variable satisfies McNaughton's Theorem
- ☐ Show that every (1 variable) function satisfying McNaughton's Theorem can be represented as the truth function of some propositional formula in 1 variable.

8.1 An Argument for Łukasiewicz's Logic

Of the three logics $\mathbb{F}^1$, $\mathbb{F}^2$ and $\mathbb{F}^3$, Łukasiewicz logic $\mathbb{F}^3$ is most well-studied.

$\mathbb{F}^3$		
$F_\neg(x)$	$=$	$1 - x$
$F_\wedge(x, y)$	$=$	$\mathsf{max}(0, x + y - 1)$
$F_\vee(x, y)$	$=$	$\mathsf{min}(1, x + y)$
$F_\rightarrow(x, y)$	$=$	$\mathsf{min}(1, 1 - x + y)$

One argument against $\mathbb{F}^1$ and $\mathbb{F}^2$ is that both fails to validate some formulas that we would otherwise expect to be true such as $F = p \vee \neg p$. This is often codified in the so-called **Law of Excluded Middle**.

(LEM) Law of Excluded Middle: $x \vee \neg x = 1$, for all $x \in [0, 1]$;

Exercise

Show that $\mathbb{F}^1$ and $\mathbb{F}^2$ fail to satisfy **(LEM)**.

And so, if we consider the **(LEM)** to be a desirable property of our logic, then we must exclude $\mathbb{F}^1$ and $\mathbb{F}^2$.

However, concerning $\mathbb{F}^1$, *if* min and max are desirable functions that we wish to express in our logic, then $\mathbb{F}^3$ is strong enough to express these already as **defined operators**. We may introduce two additional binary operators $\underline{\wedge}$ and $\underline{\vee}$ as **abbreviations** of the following formulas:

$$x \,\underline{\wedge}\, y \;=\; (x \to y) \wedge x \qquad \text{and} \qquad x \,\underline{\vee}\, y \;=\; (x \to y) \to y$$

which take on the minimum and maximum values of their arguments, respectively. This is to say that Łukasiewicz logic $\mathbb{F}^3$ is an **extension** of $\mathbb{F}^1$, and any reasoning we may like to do in regarding $\mathbb{F}^1$ can just as well be done with $\mathbb{F}^3$[1]

[1]Formally speaking, there is a translation $t : \mathit{Fml} \to \mathit{Fml}$, obtained by iteratively replacing all occurrences of $\wedge$ and $\vee$ with $\underline{\wedge}$ and $\underline{\vee}$, respectively, such that $\models^{\mathbb{F}^1} F$ if and only if $\models^{\mathbb{F}^3} t(F)$.

Exercise

Show that, for all $x, y \in [0,1]$ we have that:

$$(x \underline{\wedge} y) \;=\; \mathsf{min}(x,y) \qquad \text{and} \qquad (x \underline{\vee} y) = \;\mathsf{max}(x,y).$$

Great! So can we do the same for the product $\mathbb{F}^2$ and find some combination of formulas that would behave in $\mathbb{F}^3$ as do the $\wedge$ and $\vee$ in $\mathbb{F}^2$, that we could use as suitable abbreviation for additional defined operators?

After a short (or long) period of trial-and-error, you may convince yourself that this is, indeed, a futile and Sisyphean task, and that no combination of connectives of finite length — no matter how complex and creative — can quite mirror the behaviour of the $\mathbb{F}^2$ t-norm and t-conorm. But how can we be sure that we haven't just missed something? Maybe we just need longer formulas!

To show that this isn't possible, regardless of how creative we are, we need some sort of **characterisation result** that tells us precisely what *kinds* of functions can be expressed by formulas in $\mathbb{F}^3$. This is where McNaughton's Theorem comes in.

8.2 McNaughton's Theorem (Part 1)

When it comes to classical propositional logic, we see that every Boolean function $f : \{0,1\}^n \to \{0,1\}$ is the truth-function of *some* propositional formula. That is to say that there is some $F \in$ *Fml* such that $V(F) = f$ — for instance, we could construct a formula in disjunctive normal form by taking the disjunction of the conjunctions of literals corresponding to inputs for which $f(x_1, \ldots x_n) = 1$.

$$F \;:=\; \bigvee_{f(x_1,\ldots,x_n)=1} \left(\bigwedge_{i=1}^{n} p_i^{x_i} \right)$$

where $p_i^1 = p_i$ and $p_i^0 = \neg p_i$.

However, it should come as no surprise that the same is *not* true for arbitrary real-valued functions $f : [0,1]^n \to [0,1]$ with respect to Łukasiewicz semantics (or indeed any fuzzy semantics)[2] But then, this raises the question: just what kinds of functions *can* be represented by propositional formulas under Łukasiewicz semantics?

Theorem 8.1: McNaughton's Theorem (1 variable) [1]

A one-variable real-valued function $f : [0,1] \to [0,1]$ can be represented by a propositional formula under Łukasiewicz semantics if and only if there is some increasing sequence

$$0 = \gamma_0 < \gamma_1 < \gamma_2 < \cdots < \gamma_n = 1$$

and family of linear function $f_i(x) = a_i x + b_i$ with $a_i, b_i \in \mathbb{Z}$, for $i < n$, such that

$$f(x) = f_i(x) = a_i x + b_i \qquad \text{for all } x \in [\gamma_i, \gamma_{i+1}]$$

In particular, we have that $f_i(\gamma_{i+1}) = f_{i+1}(\gamma_{i+1})$, for all $i < n$.

In other words, McNaughton's Theorem says that we can always partition the truth function of any one-variable propositional formula is a **continuous piecewise-linear function with integer coefficients** [1]. And conversely, that *every* one-variable,

[2]A simple argument is that there there are $2^{2^{\aleph_0}}$ such functions from $[0,1]^n$ to $[0,1]$ (or $2^{\aleph_0}$ if we restrict ourselves to continuous functions), however there are 'only' $\aleph_0$ propositional formulas.

continuous piecewise-linear function with integer coefficients is the truth function of some one-variable propositional formula.

Lemma 8.2

The truth-function of every one-variable propositional formula is a continuous piecewise-linear function with integer coefficients.

Proof: We prove this by induction on the size of the formula.

- Suppose that $n = 1$ then $F = p$ is just a propositional variable, for which we can take $0 = \gamma_0 < \gamma_1 = 1$ with $\overline{F}(x) = x$, as required.

- Suppose for induction that the result holds for all formulas of length $< n$ and let φ have length n. We have four cases to consider

  - If $F = \neg G$ for some G then since G is of length $< n$ we have that there are $0 = \gamma_0 < \gamma_1 \cdots < \gamma_t = 1$ and $g_i(x) = a_i x + b_i$ for $a_i, b_i \in \mathbb{Z}$, for $i < t$, such that $\overline{G}(x) = g_i(x)$ for all $x \in [\gamma_i, \gamma_{i+1}]$.

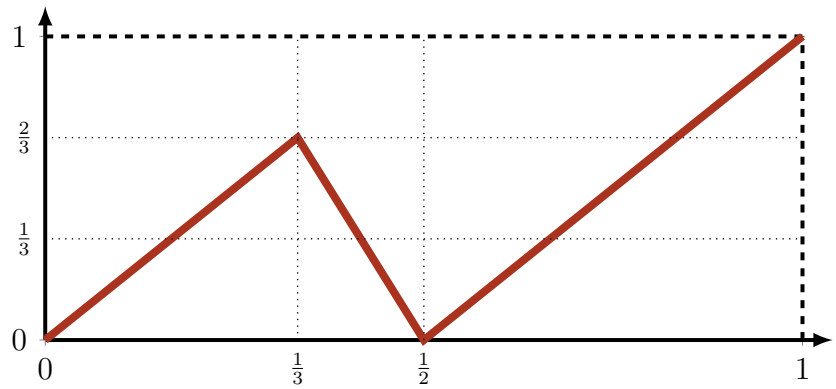
    We define a new family of functions $f_i(x) = a'_i x + b'_i$, for each $i < n$, by taking $a'_i = -a_i \in \mathbb{Z}$ and $b'_i = (1 - b_i) \in \mathbb{Z}$, so that for $x \in [\gamma_i, \gamma_{i+1}]$

$$\begin{aligned} \overline{F}(x) &= 1 - \overline{G}(x) \\ &= 1 - g_i(x) \\ &= 1 - (a_i x + b_i) \\ &= (-a_i)x + (1 - b_i) \;=\; f_i(x) \end{aligned}$$

    as required.

Exercise

Suppose that G is a formula whose truth-function $\overline{G}$ is as shown below. Sketch the truth-function of the formula $F = \neg G$ on the same axes.



- If $F = G \wedge H$ then since the length of G and H are both $< n$, by the Induction Hypothesis:

  (i) there is some sequence $0 = \gamma_0 < \gamma_1 < \cdots < \gamma_t = 0$ and $g_i(x) = a_i x + b_i$ with $a_i, b_i \in \mathbb{Z}$, such that $\overline{G}(x) = g_i(x)$ for all $x \in [\gamma_i, \gamma_{i+1}]$;

  (ii) there is some sequence $0 = \delta_0 < \delta_1 < \cdots < \delta_s = 0$ and $h_i(x) = c_i x + d_i$ with $c_i, d_i \in \mathbb{Z}$, such that $\overline{H}(x) = h_i(x)$ for all $x \in [\delta_i, \delta_{i+1}]$.

  Without loss of generality we may assume that $t = s$ and $\gamma_i = \delta_i$, for all $i < t$, since we do not require F_φ and F_ψ to not *non-linear* at each γ_i and δ_i, respectively.

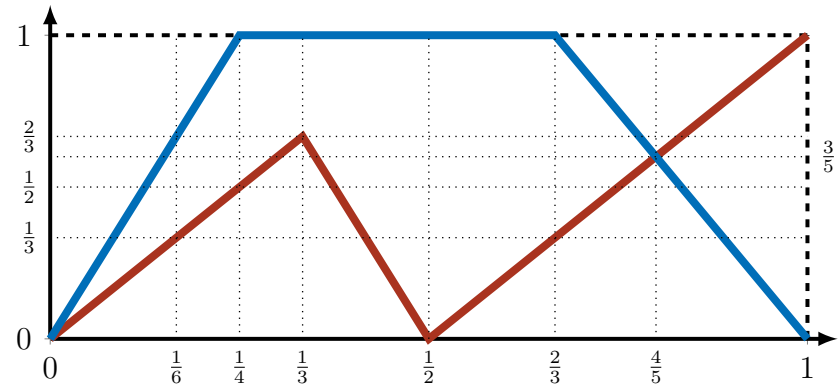
  Hence we can take $f_i(x) = (a_i + c_i)x + (b_i + d_i - 1)$ so that

$$\begin{aligned}
\overline{F}(x) &= \max(0, \overline{G}(x) + \overline{H}(x) - 1) \\
&= \max(0, (a_i x + b_i) + (c_i x + d_i) - 1) \\
&= \max(0, (a_i + c_i)x + (b_i + d_i - 1))
\end{aligned}$$

  for all $x \in [\gamma_i, \gamma_{i+1}]$, as required.

Exercise

Suppose that G and H are two formulas whose truth-function $\overline{G}$ (red) and $\overline{H}$ (blue) are as shown below. Sketch the truth-function of the formula $F = (G \wedge H)$ on the same axes.



- If $F = (G \vee H)$ then we note that, under the Łukasiewicz semantics, $F \equiv \neg(\neg G \wedge \neg H)$, and so it follows immediately from the previous two cases. Likewise, if $F = (G \to H)$, which is equivalent to $(F \vee G) \equiv \neg(F \wedge \neg G)$.

8.3 McNaughton's Theorem (Part 2)

Lemma 8.3

Let $n, m \in \mathbb{Z}$ be integers and let $f : [0,1] \to [0,1]$ be defined as

$$f_{n,m}(x) \;=\; [nx + m]_0^1$$

where $[z]_0^1 = \min(1, \max(0, z))$ denotes the restriction of z to the interval $[0,1]$. Then there exists some one-variable propositional formula $F_{n,m}$ whose truth-function is $f_{n,m}$.

Proof:

Base Case) First suppose $n = 0$, so that

$$f_{0,m}(x) = [m]_0^1 = \begin{cases} 0 & \text{if } m \leq 0 \\ 1 & \text{if } m \geq 1 \end{cases}$$

Hence, we $m \leq 0$ we can take

$$F_{0,m} = \begin{cases} \bot & \text{if } m \leq 0 \\ \top & \text{if } m \geq 1 \end{cases}$$

Where $\top = (p \vee \neg p)$ abbreviates a tautology and $\bot = (p \wedge \neg p)$ abbreviates a contradiction.

Induction Hypothesis) Suppose for all the result is true for all $n, m \in \mathbb{Z}$ where $|n| \leq k$, for some $k \geq 0$. We want to show that the result also holds when $|n| = (k+1)$.

Induction Case for $n = (k+1)$) We want to show that for all $m \in \mathbb{Z}$ there is some function $F_{n,m}$ whose truth-function is $f(x) = [nx + m]_0^1$.

To this end, let

$$f_{k,m}(x) = [kx + m]_0^1 \qquad \text{and} \qquad f_{k,(m+1)}(x) = [kx + (m+1)]_0^1$$

Then by the Induction Hypothesis, there are formulas $F_{k,m}$ and $F_{k,(m+1)}$ whose truth-functions are $f_{k,m}$ and $f_{k,(m+1)}$ respectively.

Hence, we can define a new function $F_{(k+1),m}$ by taking

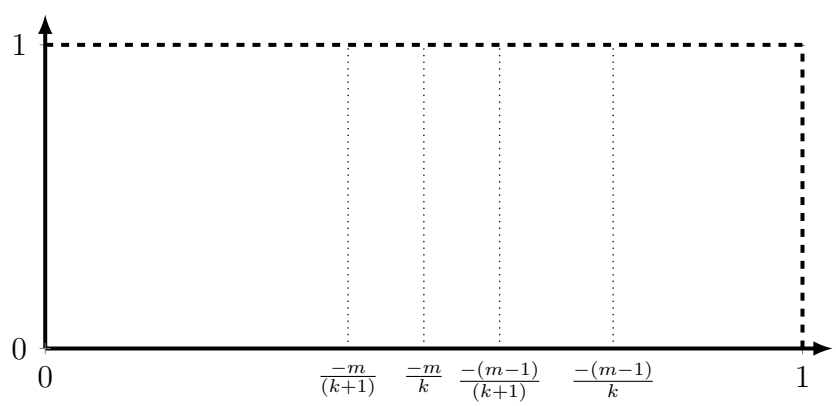
$$F_{(k+1),m} \;=\; (F_{k,m} \vee p) \wedge F_{k,(m+1)}$$

We can then verify that the truth-function of $F_{(k+1),m}$ is $f_{(k+1),m}$, by considering it's behaviour over the intervals $[0, \frac{-m}{(k+1)}]$, $[\frac{-m}{(k+1)}, \frac{-m}{k}]$, $[\frac{-m}{k}, \frac{(1-m)}{(k+1)}]$ and $[\frac{(1-m)}{(k+1)}, 1]$.

- Consider first the truth functions for $F_{k,m}$ and p:

Notes

Sketch on the following axes, the truth functions of $F_{k,m}$ and p.

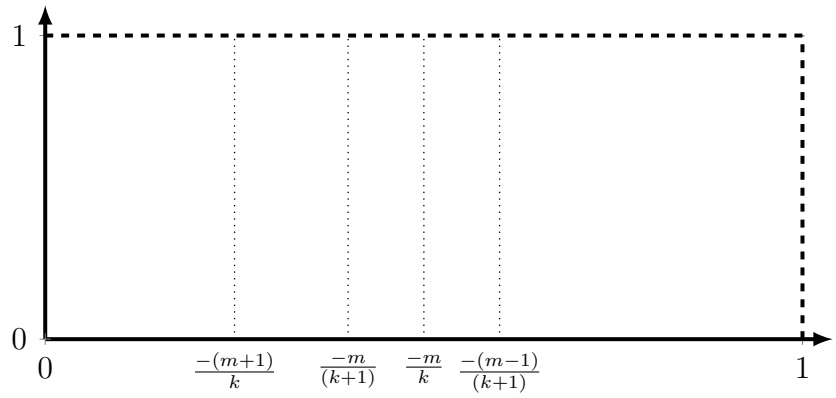


Over the interval $[0, \frac{-m}{k}]$ we have that $f_{k,m}(x) + x = 0 + x = x$, while over the interval $[\frac{-(m-1)}{(k+1)}, 1]$ we have that $f_{k,m}(x) + x \geq 1$. Over the interval $[\frac{-m}{k}, \frac{-(m-1)}{(k+1)}]$ we have that $f_{k,m}(x) + x = (kx + m) + x = (k+1)x + m$.

- We can then consider the truth functions of $(F_{k,m} \vee p)$ and $F_{k,(m-1)}$:

Notes

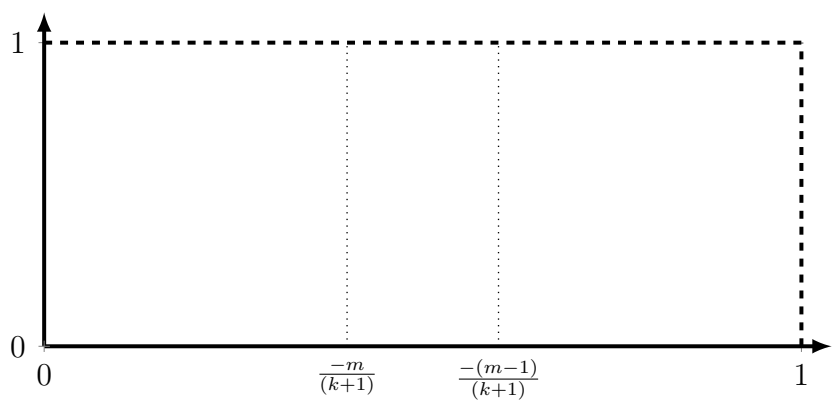
Sketch on the following axes, the truth functions of $(F_{k,m} \vee p)$ and $F_{k,(m+1)}$:



Over the interval $[0, \frac{-m}{(k+1)}]$ we have that $x + f_{k,(m+1)}(x) - 1 \leq 0$, while over the interval $[\frac{-m}{k}, 1]$ we have that $f_{k,(m+1)}(x) = 1$ and so $g(x) + fk, (m+1) - 1 = g(x)$. Over the interval $[\frac{-m}{(k+1)}, \frac{-m}{k}]$ we have that $x + f_{k,(m+1)}(x) - 1 = x + (kx + (m+1)) - 1 = (k+1)x + m$.

Notes

Sketch on the following axes, the truth function of $(F_{k,m} \vee p) \wedge F_{k,(m+1)}$.



Induction Case for $n = -(k+1)$) We want to show that for all $m \in \mathbb{Z}$ there is some function $F_{n,m}$ whose truth-function is $f(x) = [nx + m]_0^1$.

To this end, let

$$f_{-k,m}(x) = [-kx + m]_0^1 \qquad \text{and} \qquad f_{-k,(m-1)}(x) = [-kx + (m-1)]_0^1$$

As above, by the Induction Hypothesis, there are formulas $F_{-k,m}$ and $F_{-k,(m-1)}$ whose truth-functions are $f_{-k,m}$ and $f_{-k,(m-1)}$ respectively.

Hence, we can define a new function $F_{-(k+1),m}$ by taking

$$F_{-(k+1),m} \;=\; (F_{-k,(m-1)} \vee \neg p) \wedge F_{-k,m}$$

We can then verify that the truth-function of $F_{-(k+1),m}$ is $f_{-(k+1),m}$, by considering it's behaviour over the intervals $[0, \frac{-m}{(k+1)}]$, $[\frac{-m}{(k+1)}, \frac{-m}{k}]$, $[\frac{-m}{k}, \frac{(1-m)}{(k+1)}]$ and $[\frac{(1-m)}{(k+1)}, 1]$.

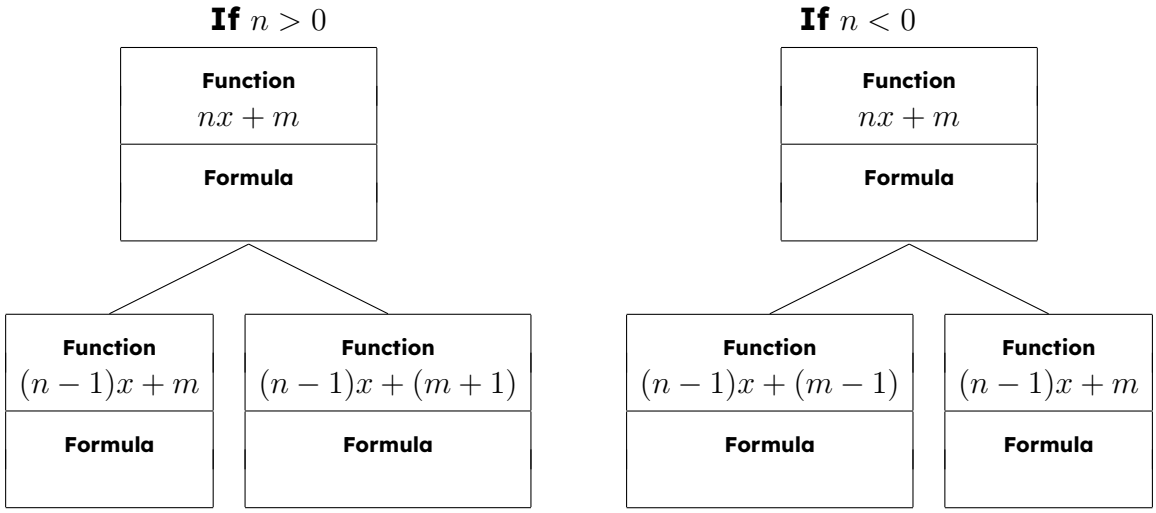
Hence it follows by induction that for all $n, m \in \mathbb{Z}$ there is some formula $F_{n,m}$ whose truth function is $f_{n,m}(x) = [nx + m]_0^1$, as required. ∎

We can turn the induction of Lemma **??** into a algorithm for recursively generating formulas whose truth-function is of the form $f(x) = [nx+m]_0^1$ for any integers $n, m \in \mathbb{Z}$, as follows:

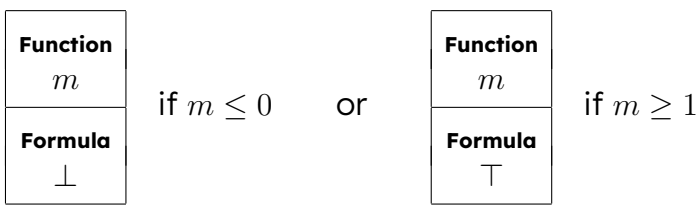
Step 1) Start at the root of a tree with a node labelled with

Function
$nx + m$
Formula

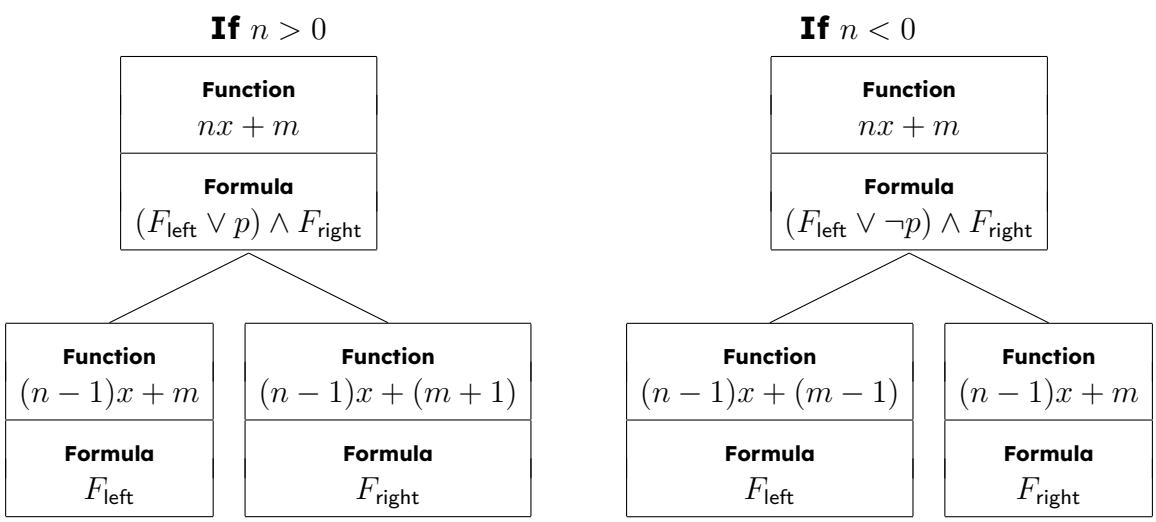
Step 2) From each node of the tree, if $n \neq 0$ then create two child nodes, each labelled as follows, depending on whether n if positive or negative:



Step 3) Once we reach a node with $n = 0$ we can fill in the cell beneath with a corresponding propositional formula



Step 4) Working our way up the tree to the root, we can populate the formula cell based on the formulas contained in the formula cells of the



We can **simplify** any formulas as we go by applying the rules that

$$
\begin{aligned}
(F \wedge \bot) &\equiv (\bot \wedge F) \equiv \bot \\
(F \wedge \top) &\equiv (\top \wedge F) \equiv F \\
(F \vee \bot) &\equiv (\bot \vee F) \equiv F \\
(F \vee \top) &\equiv (\top \vee F) \equiv \top
\end{aligned}
$$

Exercise

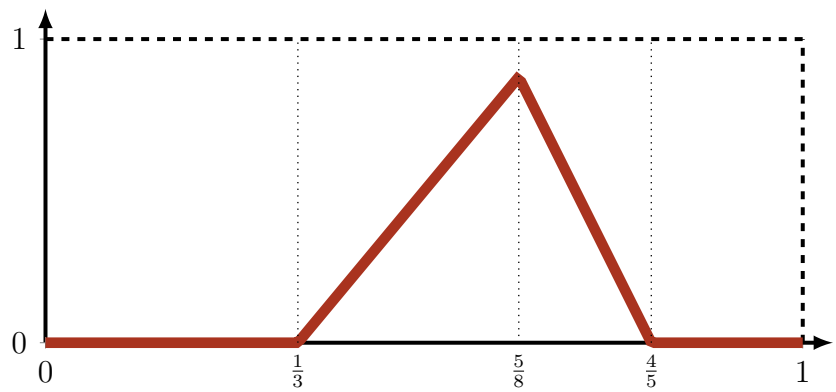
Use the tree algorithm to construct a propositional formula whose truth function is

$$f(x) \;=\; [3x-1]_0^1$$

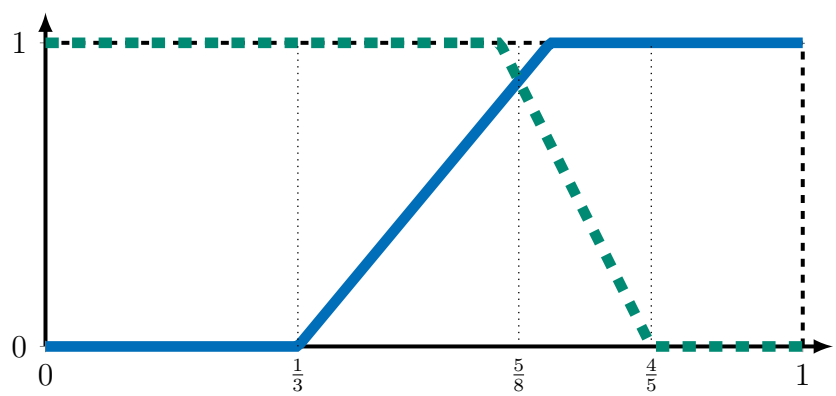
Lemma 8.3 shows that for any (continuous) piecewise linear function f, we can find propositional formulas $F_1, \ldots, F_n$ such that truth-function of F_i aligns with f along the interval $[\gamma_{i-1}, \gamma_i]$. However, we still need to a way to combine these formulas into

a single formula that aligns with f along the whole of its domain $[0,1]$. We achieve this with some judicious combination of our defined operators $\underline{\wedge}$ (min) and $\underline{\vee}$ (max). For example, consider the following function

$$f(x) = \begin{cases} 0 & \text{if } x \leq \frac{1}{3} \\ 3x - 1 & \text{if } \frac{1}{2} < x \leq \frac{5}{8} \\ 4 - 5x & \text{if } \frac{5}{8} < x \leq \frac{4}{5} \\ 0 & \text{if } x > \frac{4}{5} \end{cases}$$



Then it is clear that f is simply takes on the minimum of $f_{3,-1} = [3x-1]_0^1$ (blue) and $f_{-5,4} = [-5x+4]_0^1$ (green/dashed), shown below:

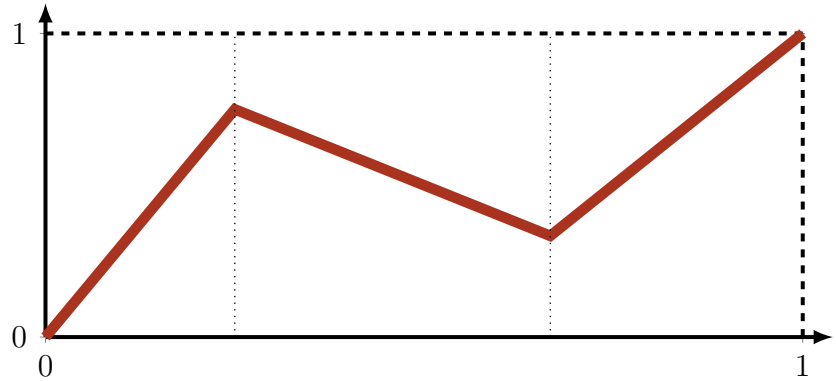


Hence we would be able to take F to be

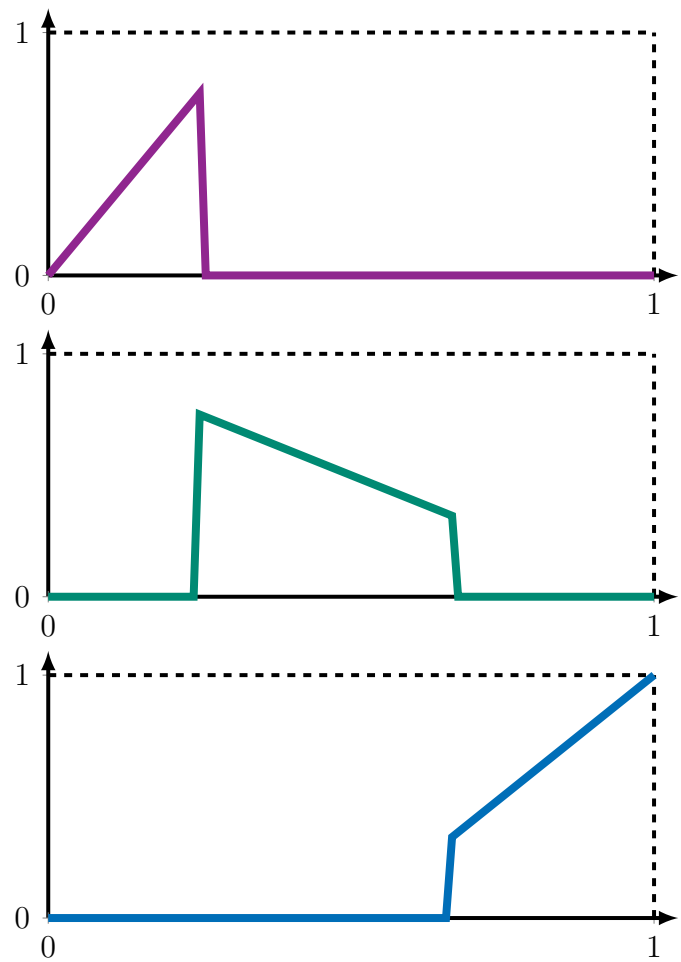
$$F = F_{3,-1} \underline{\wedge} F_{-5,4}$$

where $F_{3,-1}$ and $F_{-5,4}$ are as defined in the proof of Lemma 8.3.

For more complex functions such as the following



we cannot always simply take a minimum or maximum of functions of the form $f_{n,m}$. Instead, what we must first do, is decompose f into 'wedges-shaped' pieces like so:



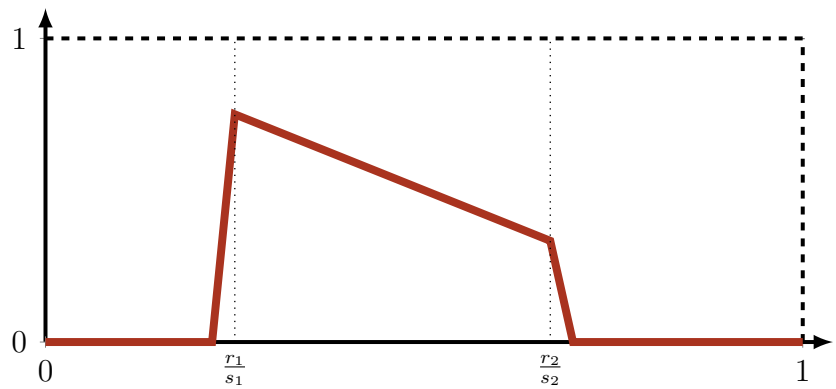
each of which can be described by a piecewise-linear function with integers coefficients, and whose maximum is the desired function f. (Note that the 'vertical' portions of

these functions are not actually vertical, as that would not be a well-defined function, but rather linear functions with a large positive or negative gradient.)

8.4 McNaughton's Theorem (Part 3) — Optional

[This section is for the sake of completeness and is not examinable]

We want to show that for any linear function $f_{n,m}(x) = [nx + m]_0^1$ with $n, m \in \mathbb{Z}$, and rational points $0 \leq \frac{r_1}{s_1} < \frac{r_2}{s_2} \leq 1$, we can construct a 'wedge-shaped' function:



in which the slope slope of the linear components immediately before r_1/s_2 and immediately after r_2/s_2 can be made arbitrarily steep.

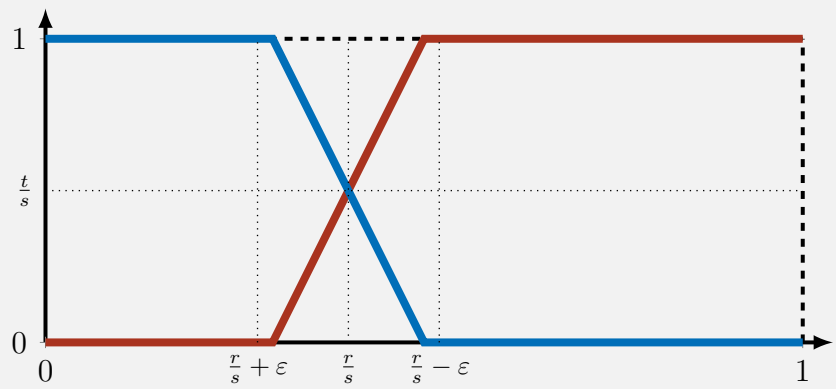
Lemma 8.4

Let $r, s, t \in \mathbb{N}$ be such that gcd$(r, s) = 1$. Then for all $\varepsilon > 0$, there exists formulas G_1 and G_2, with truth-functions g_1 and g_2, respectively, such that:

(i) $g_i(\frac{r}{s}) = \frac{t}{s}$, for $i = 1, 2$,

(ii) $g_1(x) = 0$ and $g_2(x) = 1$, for all $x < \frac{r}{s} - \varepsilon$,

(iii) $g_1(x) = 1$ and $g_2(x) = 0$, for all $x > \frac{r}{s} + \varepsilon$,



Proof: Let r, s, t be as described, and let $\varepsilon > 0$ be fixed. By **Bézout's Identity**, we have that there are integers $a, b \in \mathbb{Z}$ such that $ar + bs = \gcd(r, s) = 1$, and hence $art + bst = t$ which is to say that

$$\begin{aligned} ar + bs = 1 \quad &\Longrightarrow \quad art + bst = t \\ &\Longrightarrow \quad art + bst + (krs - krs) = t \qquad \text{for any } k \in \mathbb{Z} \\ &\Longrightarrow \quad (at + ks)r + (bt - kr)s = t \end{aligned}$$

Hence, we can take $n = (at + ks)$ and $m = (bt - kr)$ so that

$$f_{n,m}\left(\frac{r}{s}\right) = n\left(\frac{r}{s}\right) + m = (at + ks)\left(\frac{r}{s}\right) + (bt - kr) = \left(\frac{t}{s}\right)$$

By choosing k to be sufficiently large and positive, we can ensure that $f_{n,m}(x) = [nx + m]_0^1 = 0$ for all $x < \frac{r}{s} - \varepsilon$, as required for g_1. And by choosing k to be sufficiently large and negative, we can ensure that $f_{n,m}(x) = [nx + m]_0^1 = 0$ for all $x > \frac{r}{s} + \varepsilon$, as required for g_2. ∎

With Lemma 8.4, we are able to construct a formula corresponding to any 'wedge-shape' function as shown above, which then allows us to complete the proof of McNaughton's Theorem in one-variable (Theorem 8.2).

Proof:

- Let piecewise linear function $f : [0,1] \to [0,1]$ be a continuous piecewise linear function with $0 = \gamma_0 < \gamma_1 < \cdots < \gamma_n = 1$, which that $f(x) = a_i(x) + b_i$, for all $x \in [\gamma_i, \gamma_i + 1]$.

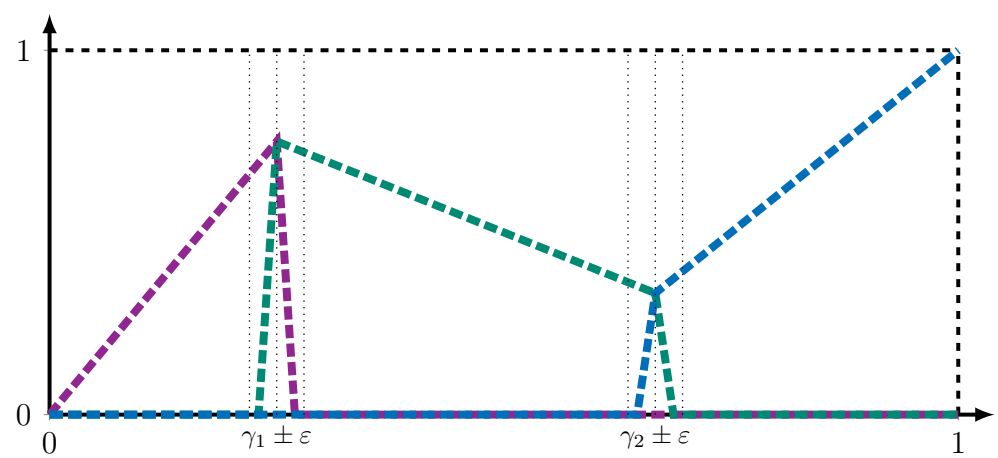
- For each $i < n$, let

$$F_i = (G_1^i \mathbin{\underline{\wedge}} F_{a_i,b_i} \mathbin{\underline{\wedge}} G_2^i)$$

  where G_1^i and G_2^i denote the formulas corresponding to the functions g_1 and g_2 of Lemma 8.4 such that (i) $g_1(\gamma_i) = a_i\gamma_i + b_{i+1}$ and $g_2(\gamma_{i+1}) = a_{i+1}\gamma_{i+1} + b_{i+1}$, and (ii) ε is chosen to be sufficiently small so that g_1 and g_2 not not intersect f at any other point $x \in [0,1]$.

- We then define F to be the maximum of each of the formulas F_i, for $i < n$, corresponding to each of the wedge shaped sections:

$$F \;=\; F_0 \mathbin{\underline{\vee}} F_1 \mathbin{\underline{\vee}} F_2 \mathbin{\underline{\vee}} \cdots \mathbin{\underline{\vee}} F_n$$



References

[1] Robert McNaughton. A theorem about infinite-valued sentential logic. *The Journal of Symbolic Logic*, 16(1):1–13, 1951.

8.5 Reflections

Reflections

Reflect and summarise the key points from this week's content.